Melting Temperature Estimation: Wallace Rule vs Nearest-Neighbor Model

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1 Introduction

The melting temperature (T_m) can be defined as the temperature at which 50% of a DNA sequence is double-stranded (dsDNA), while the other 50% is single-stranded (ssDNA).

In molecular assays, measuring the right T_m is crucial because it helps determine the hybridization temperature (often 3-5 °C below T_m). There have been different methods to determine T_m. Simpler methods consider only the content of bases in a sequence, such as the Wallace rule. However, there are many variables that affect T_m, which are considered in the Nearest-Neighbor model (NN), currently the standard for determining T_m.

This small analysis, through simulations, focused on exploring the differences in the T_m calculation between the Wallace rule and NN. It was expected that the NN model would provide more accurate estimates. Still, one goal was to see if the Wallace rule would match in precision in a given situation.

2 Methods

The simulations were developed with the R programming language (in the RStudio environment). First, three groups of primers were simulated: small (12-17 bp), medium (18-24 bp), and large (25-30 bp) (Table 1). For each group, 1000 primers were generated (Table 1).

Group	Length (bp)	Simulated primers
Small	12–17	1000
Medium	18–24	1000
Large	25–30	1000

Table 1: Primer groups used for simulation

The first method to calculate the T_m was the Wallace rule:

$$T_m = 4(G + C) + 2(A + T)$$

This rule considers only the absolute frequency of the bases in a sequence.

The second method to determine T_m was NN with salt correction:

$$T_m = \frac{\Delta H^\circ}{\Delta S^\circ + R \ln\left(\frac{C_t}{4}\right)} - 273.15 + 16.6 \log_{10}[\mathrm{Na}^+]$$

This formula relies on thermodynamic parameters derived from the nearest-neighbor model, which considers the dinucleotide composition of the DNA sequence (SantaLucia et al., 1996). The ΔH° and ΔS° values were obtained from a unified parameter set (SantaLucia, 1998), and the effect of salt concentration on T_m was incorporated using an empirical correction (Schildkraut & Lifson, 1965). Primer and salt concentrations were set to 0.25 μM and 50 mM, respectively.

After determining T_m for the three primer groups with both methods, the difference in T_m (ΔT_m) and $|\Delta T_m|$ were calculated.

3 Results

The increase in $|\Delta T_m|$ is proportional to the increase in primer length (Figure 1). The large primer group had higher values of $|\Delta T_m|$, the lowest $|\Delta T_m|$ values were observed in the small primer group (Figure 1).

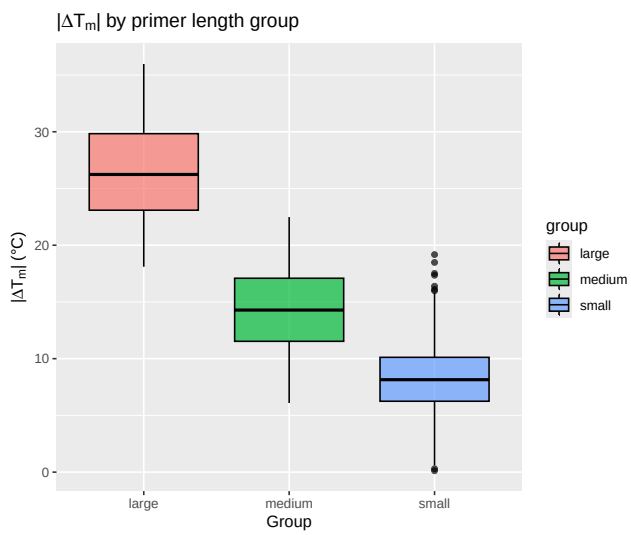

Figure 1: Absolute difference in T_m ($|\Delta T_m|$) by primer length group

In all of the primer groups, the T_m calculated by NN was generally lower than the Wallace rule output (Figure 2). However, values of both methods became closer when the T_m was around 50 °C (Figure 2).

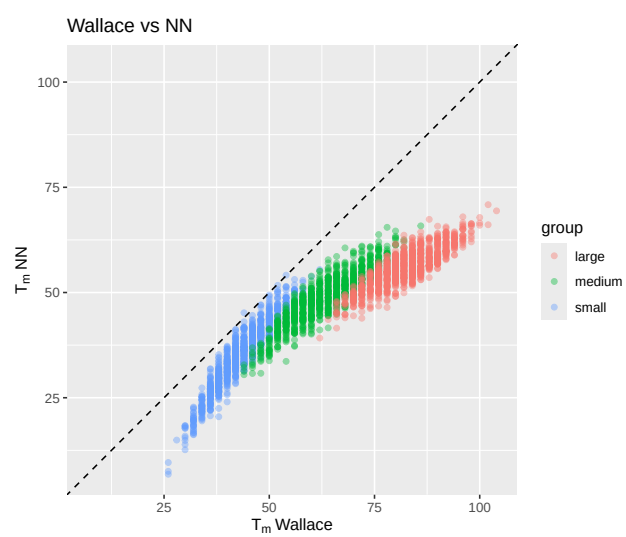

Figure 2: Wallace vs NN T_m values

In none of the primer groups was ΔT_m centered around 0 (Figure 3). Only in

the small primer group were a few values observed close to 0 (Figure 3).

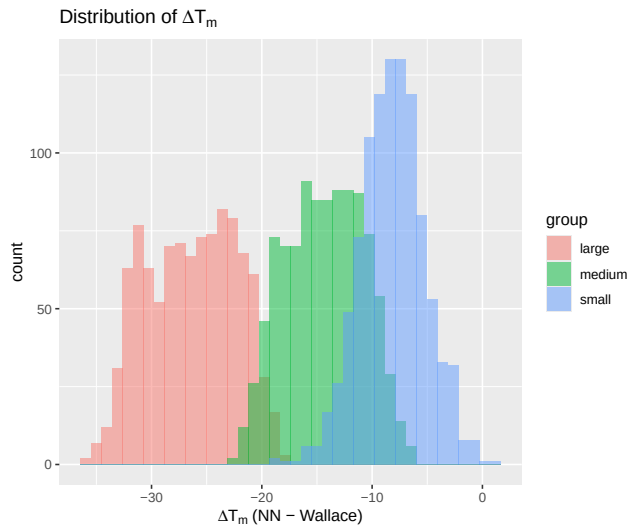

Figure 3: Distribution of ΔT_m

The large primer group showed a mean $|\Delta T_m|$ of 26.38 with a 95% confidence interval (CI) of (26.13, 26.63) (Table 2). Similarly, the medium and small primer groups had mean values of 14.31 (CI: 14.09, 14.53) and 8.14 (CI: 7.95, 8.32), respectively (Table 2). None of the confidence intervals include 0, indicating that the difference between the two methods is consistently different from zero across all primer length groups.

Group	$\lvert\Delta T_m\rvert$ Mean	$\lvert\Delta T_m\rvert$ 95% CI
Large	26.38	(26.13, 26.63)
Medium	14.31	(14.09, 14.53)
Small	8.14	(7.95, 8.32)

Table 2: Mean and 95% confidence interval of $|\Delta T_m|$ for each primer group

4 Conclusion

The Wallace rule does not seem to get close to the NN method in any length of primers. Also, the Wallace rule seems to overestimate T_m in every situation. As a formula for educational purposes, it would be useful. However, it does not seem to be a reliable method for accurate T_m estimation. Also, this analysis does not consider factors such as Mg^{2+} concentration or secondary structure formation, which may also affect T_m.

5 References

- SantaLucia, J., Jr., Allawi, H. T., & Seneviratne, P. A. (1996). Improved nearest-neighbor parameters for predicting DNA duplex stability. *Biochemistry*, 35(11), 3555–3562. https://doi.org/10.1021/bi951907q
- SantaLucia J., Jr (1998). A unified view of polymer, dumbbell, and oligonucleotide DNA nearest-neighbor thermodynamics. *Proceedings of the National Academy of Sciences*, 95(4), 1460–1465. https://doi.org/10.1073/pnas.95.4.1460
- Schildkraut, C., & Lifson, S. (1965). Dependence of the melting temperature of DNA on salt concentration. *Biopolymers*, 3(2), 195–208. https://doi.org/10.1002/bip.360030207